

Bowen Yan

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Research Interests	Reasoning and Agents; Failure-Driven Self-Improvement; Embodied AI	
Education	Tsinghua University	2021–2024
	M.S. in Electrical Engineering(Robotics & Intelligent Systems)	
	Hunan University	2016–2020
	B.E. in Mechanical Engineering	
Research Appointments	Shanghai Artificial Intelligence Laboratory (Shanghai AI Lab)	2025–Present
	Researcher	
	Tsinghua University	2024–2025
	Research Assistant	
Publications & Manuscripts	[1] Can VLMs Diagnose and Recover from VLA Manipulation Faults?	
	Bowen Yan[†] , Jiahao Xiao [†] , Kehui Liu, Jianbo Zhang, Zicheng Zhang, Qi Jia, Zhongjie Jia, Haoming Song, Chunyi Li, Bin Zhao, Guangtao Zhai.	
	ICML 2026.	
	Develops a VLM-guided recovery framework that diagnoses VLA execution failures and translates fault understanding into spatial coordinate corrections and temporal rollback decisions, generating corrective actions that improve manipulation success.	
	[2] TextMani: Diagnosing and Repairing Constraint Formalization in LLM-Based Robotic Manipulation	
	Bowen Yan[†] , Zhongjie Jia [†] , Chunyi Li, Bin Zhao, Guangtao Zhai.	
	<i>Under Review at ICLR 2027.</i>	
	Builds an end-to-end LLM pipeline from natural-language task understanding to explicit physical constraint reasoning, executable action generation, and execution-grounded failure diagnosis and localized repair.	
	[3] ActBox: Action Boxes as Visual Communication for VLM Understanding and Control	
	Bowen Yan , Chunyi Li, Guangtao Zhai.	
	<i>Under Review at ICASSP 2027.</i>	
	Investigates how spatial communication modalities—including visual boxes, image grids, and textual coordinates—enhance VLM contextual understanding and improve the accuracy of directly generated task-control actions.	
	[4] From Statistical to Pedagogical View: A Learner-Centric Perspective on Embodied Data	
	Jiahao Xiao [†] , Bowen Yan[†] , Jianbo Zhang [†] , Jia Wang, Zicheng Zhang, Yue Shi, Yuan Tian, Zhengxue Cheng, Chunyi Li, Guangtao Zhai.	
	<i>Under Major Revision at npj Artificial Intelligence (Embodied AI Collection).</i>	
	Introduces Learning Effectiveness of Data (LED), a learner-centric metric that predicts policy-learning effectiveness and guides efficient embodied data curation, enabling stronger policy learning with substantially less data.	
	[5] Puff-Pssss: Design and Optimization of a Biomimetic Soft Robot for Adaptive Pipe Inspection	
	Bowen Yan[†] , Xinyu Yu [†] , Qian Zhou, Xueqian Wang.	
	<i>Under Major Revision at IEEE Robotics and Automation Letters (RA-L).</i>	
	Developed a complete soft robotic system integrating mechanical structures, custom soft actuators, optical sensing hardware, and perception algorithms, with joint optimization of actuation and sensing parameters through finite element analysis and system-level experiments.	

[†] Equal contribution.